



RAMCO INSTITUTE OF TECHNOLOGY

(An Autonomous Institution)

Approved by AICTE, New Delhi & Affiliated to Anna University
NAAC Accredited with 'A+' Grade & An ISO 9001: 2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE, MECH and CIVIL

Department of Computer Science and Engineering

Academic Year 2025 – 2026 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. CSE

Course Code & Title: CS3452 – Theory of Computation

Name of the Faculty Member(s): Mrs. S. Manjula

Innovative Practice Description

Unit / Topic: Unit I / Equivalence of NFAs with and without ϵ -moves

Course Outcome: CO 1

Topic Learning Outcome: TLO 2

Activity Chosen: Collaborative Learning

Date of Activity: 09.01.2026

Venue: CSE Seminar Hall

Justification:

- Collaborative learning is an educational approach to teaching and learning that involves groups of students working together to solve problems, complete tasks, or learn new concepts. It promotes active engagement and deeper understanding of abstract computational topics.
- By collaborative learning, students gained in-depth knowledge of the equivalence of NFAs with and without ϵ -moves, making the concepts more accessible and engaging.
- The activity encouraged students to discuss, debate, and verify each other's understanding, which reinforced conceptual clarity and improved retention.
- It helped students make the concepts more interesting and enabled them to go beyond the regular syllabus through peer interaction.

Time Allotted for the Activity: 30 Minutes

Details of the Implementation:

The activity was conducted during a scheduled class hour in the CSE Seminar Hall with IV Semester B.E. CSE students as part of the Collaborative Learning initiative. The focus of the session was on the "Equivalence of NFAs with and without ϵ -moves", a key topic in Theory of Computation. The implementation involved the following steps:

Step 1: Group Formation

- Students were divided into 13 groups, with each group consisting of 4 to 5 members.
- Each group was assigned a unique problem statement related to the equivalence of NFAs with and without ϵ -transitions.

Step 2: Problem Solving

- Each group worked collaboratively to discuss, analyze, and solve their assigned problem.
- Students applied their understanding of ϵ -closure, NFA construction, and transition tables to arrive at solutions.

Step 3: Verification and Feedback

- The instructor circulated among the groups, checked the solutions prepared by each team, and verified the correctness of the answers.
- Immediate feedback was provided to each group to address misconceptions and reinforce correct understanding.

Step 4: Consolidation

- After all groups completed their tasks, the faculty member consolidated the key points discussed across all groups.
- This assisted students in recalling the topic, generating new ideas, and answering any remaining doubts about NFA equivalence.

CO – PO / PSO Mapping:

CO	PO1	PO2	PO3	PO4	PO9	PO10	PSO1
CO 1	2	2	2	1	1	1	1

1–Low 2–Moderate 3–High

PO / PSO Mapped:

Innovative Practice	PO1	PO2	PO3	PO4	PO9	PO10	PSO1
Collaborative Learning – Equivalence of NFAs with and without ϵ -moves	2	2	2	1	1	1	1

PO / PSO Justification:

PO / PSO	Justification
PO1	Students applied mathematical and engineering concepts by constructing and analysing NFAs with and without ϵ -moves.
PO2	Students identified and solved NFA equivalence problems using formal language theory concepts.
PO3	Students designed finite automata with and without ϵ -transitions to satisfy language recognition requirements.
PO4	Students prepared transition tables, analysed state transitions, and verified NFA equivalence.
PO9	Students worked effectively in groups to solve problems and verify answers together.
PO10	Students communicated clearly by explaining transition steps and presenting their solutions.
PO12	Students understood the importance of finite automata theory in modern computing and explored the concepts independently.

PSO1

By designing the FA, students will be able to develop software components that recognize appropriate regular expressions.

Images / Screenshot of the Practice:



Fig 1: Faculty monitoring groups during the collaborative learning activity



Fig 2: Activity Overview during the session

Reflective Critique:**Feedback of Practice from Students and Other Stakeholders:**

- Students expressed that the collaborative activity helped them identify their own understanding level of NFA equivalence concepts.
- Students informed the instructor that working in groups motivated them to be more attentive in class and to raise doubts effectively.
- Positive feedback was received from students who appreciated the hands-on, group-based approach to solving NFA conversion problems.

Benefit of the Practice:

- The activity helped the instructor analyse, evaluate, and improve students' learning through real-time observation and feedback.
- Students were able to explain NFA equivalence concepts confidently without confusion after the session.
- The collaborative approach provided greater clarity on the topic and enhanced peer-to-peer knowledge transfer.
- Outcome attainment improved as students engaged more deeply with the formal language concepts through group problem solving.

Challenges Faced in Implementation:

- A few students were initially hesitant to actively participate in group discussions and problem solving.
- These students were motivated by highlighting the personal and academic benefits of collaborative learning and active participation.

References:

- <https://www.edsys.in/what-is-peer-teaching/>
- <https://www.opencolleges.edu.au/informed/features/peer-teaching/>
- <https://tilt.colostate.edu/TipsAndGuides/Tip/180>
- <https://www.gdrc.org/kmgmt/c-learn/>
- <https://www.edutopia.org/topic/collaborative-learning>

Signature of Faculty Member**HOD / CSE**